

ANALYSIS OF THE CASH FLOW STATEMENT: RATIOS AND CHARTS

Chapter 10



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- ☐ INTRODUCTION
- ☐ *CASH FLOW BASED* RATIOS
- ☐ QUALITY OF EARNINGS RATIO
- ☐ THE CFOA – CAPEX VERSION OF FREE CASH FLOW
- ☐ GEORGIA TECH FINANCIAL ANALYSIS LAB, EARNINGS QUALITY TRENDS: JANUARY 2013
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Analysts increased their attention on cash flows, for many reasons:

- ☐ **“Cash is King”** (end of 80s): a company’s net income must convert, sooner or later, into cash
- ☐ Common use of **cash flow-based valuations** (DCF and EVA)
- ☐ Earnings are subject to **aggressive accounting policies** (earnings management / earnings quality)

One might hear such statements as:

- ☐ *“Cash flow is real and not subject to the vagaries of GAAP or the whims of the accountants”*
- ☐ *“It’s virtually impossible to manage cash flows”*
- ☐ *“Profit is an opinion. Cash is a fact”*
- ☐ *“Dividends are tangible cash, and therefore impossible to fake”*

By using both *standard* and *reformulated* cash flow statement, it is possible to calculate some interesting ratios, based on cash flow measures.

These **cash flow-based ratios are not standardized: you can find different calculation approaches for similar ratios**

Many cash flow based ratios are “coverage” ratios: to verify if a positive cash flow (in the numerator) is higher than some negative cash flows (in the denominator) so that you expect a ratio higher than 1.

These ratios focus our attention on “credit analysis”

OCF: Operating Cash Flow
CFOA: Cash Flow from Operating Activities

Target Principal on Debts Repayments	=	$\frac{\text{OCF (or EBITDA)}}{\text{Interest exp. + Principal Payments}}$	How the operating results are able to cover/support the cash flows that are required to pay the borrowers (interest and reimbursements). It is substantially a Debt Service Coverage Ratio (DSCR)
Cash Interest Coverage	=	$\frac{\text{OCF (or EBITDA)}}{\text{Interests paid}}$	The interest coverage ratio is a measure of the number of times a company could make the interest payments on its debt with its Operating Cash Flow
Ability to pay Obligations (or Cash Debt Coverage)	=	$\frac{\text{OCF or CFOA}}{\text{Total Liabilities}}$	This ratio shows the percentage of debts that the operating cash flow (or cash flow from operating activities) can repay (you may use financial debts instead of total liabilities); if you reverse this ratios the result is a number of years the company needs to pay all the debts
Cash Flow Ratio	=	$\frac{\text{CFOA}}{\text{Short-Term Liabilities}}$	This “liquidity” ratio shows the percentage of short-term debts that current cash flow from operating activities can repay; if you reverse this ratios the result is a number of years the company needs to pay all the s-t debts
Long Term Repayment Ratio	=	$\frac{\text{Long Term Debt paid}}{\text{CFOA}}$	A “solidity” ratio that measures the portion of the cash flow from operating activities that is absorbed by the repayment of long-term debts (you may also reverse this ratio)



This ratio is used ONLY to show a dramatic cash shortage (financial distress)

$$\text{Defensive Interval Ratio (Period) or Cash Days Balance} = \frac{\text{Cash \& Equivalents (+ A/R)}}{\text{"cash" Operating Expenses / 360 (avg. daily cash expenses)}}$$

A "solvency" ratio that measures how many days a company can operate without having to access non-cash assets (this ratio is normally 1-2 weeks for companies that are going bankruptcy....); in the numerator you may exclude the A/R because if the company is close to bankruptcy there is no time to collect the receivables...; in the denominator better to use the "cash" operating expenses (excluding non monetary items, ex. depreciation)



Many analysts used to calculate this ratio to show that airlines companies were in a dramatic financial distress. But they had to adjust the ratio because they observed that if an airline in crisis still operates:

- every day it generates revenues
- but at the same time it is not able to cover the daily operating expenses (staff, jet fuel etc.).

So the ratio was adjusted as:

$$\frac{\text{Cash \& Equivalents}}{\text{Daily Operating Loss}}$$

Ratios that use cash flows for comparisons similar to other "traditional ratios"

$$\text{Cash Profitability} = \frac{\text{OCF or CFOA}}{\text{Sales Revenue}}$$

This ratio measure the cash flow generated from sales (similar to Return on Sales and Profit Margin)

$$\text{Cash Return on Assets} = \frac{\text{OCF or CFOA}}{\text{Average Total Assets}}$$

This ratio indicates the percentage of cash generated from using total assets (similar to ROIC or ROA). This ratio directly measures the cash flows from operating activities generated from the company's assets



Ratios that compare cash flows from operating, investing and financing sections

Cash Flow Adequacy	=	$\frac{\text{CFOA}}{\text{Fixed assets purchased + Long term debt paid + cash Dividends distributed}}$	A cash sufficiency ratio which measures an entity's ability to replenish its productive base (CAPEX), pay long-term debt, and make distributions to owners
Funds Flow Coverage	=	$\frac{\text{OCF (or EBITDA)}}{\text{Interest exp. + Debt repayments / (1-t) + Preference dividend / (1-t)}}$	How the operating results are able to cover/support the cash flows that are required for "fixed" payments to borrowers and shareholders (interest and reimbursements and preference dividends)
Free Cash Flow Ratio	=	$\frac{\text{Operating Cash Flow}}{\text{Capital Expenditures}}$	It measure the degree to which the Operating Cash Flow covers the CAPEX (it is higher than 1 when the Operating Free cash Flow is positive)

Ratios that analyze the cash "meaning" of the Net Income

Cash Earnings	=	$\frac{\text{Net Income + Non Monetary Items (Depreciation, amortization and impairment)}}{\text{CFOA}}$	It measure the potential <i>cash flow from operating activities</i> before any change in working capital items
Cash Flow – Earnings Index	=	$\frac{\text{CFOA}}{\text{Net Income (EAT)}}$	This ratio may indicate low quality earnings from aggressive accounting or accounting irregularities (because Cash Earnings is > Net Income, this ratio should be > 1 if the change in net working capital is 0).

Let's see more on the next slides



**Cash Flow from
Operating Activities**

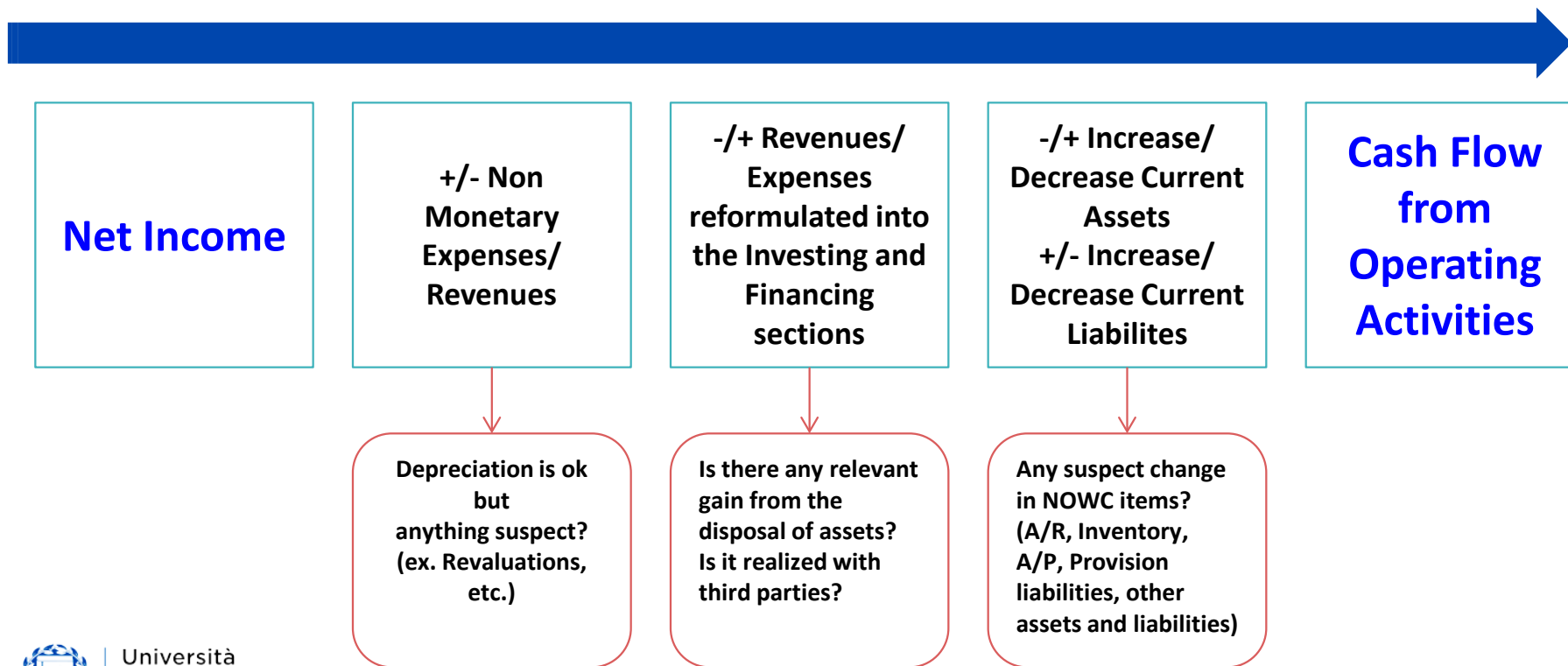
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Net Income

It measures the difference between the cash flow from operations and the net income.

If significantly lower than 1 (ex. 0,5-0,6), it may reveal the existence of aggressive accounting policies than lead to an overstatement of net income, based on accounting estimates manipulations and/or other frauds (cut-off, unrecorded liabilities etc.).

This ratio is the starting point for **Earnings Quality and Earnings Management** research studies

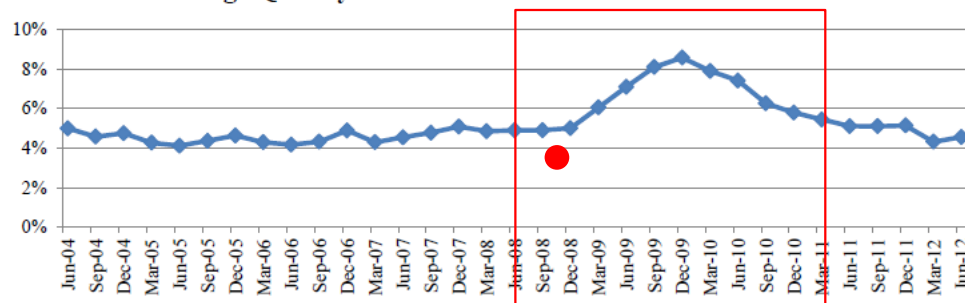


- There is an alternative Free Cash Flow: **CFOA – CAPEX** (rather than OCF – CAPEX), where CFOA is measured with the indirect method starting from the net income (and not from the operating income): it measures the **amount of liquidity created or destroyed after operating cash flow has covered interest, taxes and CAPEX**.
- E. Ketz (*Free Cash Flow and Business Combinations*, The CPA Journal, November 2016) argues this is **more comparable to net income than operating cash flow** alone, because net income already carries a subtraction for depreciation, whereas CFOA adds it back; **subtracting CAPEX restores an analogous charge for the use of long-lived assets**, since depreciation charges — and the concomitant gain or loss on disposal — ultimately equal the firm's cash flows on the asset being depreciated.
- The consequence is that **this Free Cash Flow depends on the ratio CAPEX / D&A** — the productive-capacity replacement ratio. When that ratio is 1, the add-back of depreciation and the subtraction of CAPEX offset exactly.
- So, the **Quality of Earnings Ratio** can be calculated as:

$$\frac{\text{CFOA} - \text{CAPEX}}{\text{NET INCOME}}$$



Earnings Quality Indicator - All Non-financial Industries



● October 2008: Start of the Financial Crisis

A Return to "Normal" Levels

In this study, we look at the trend in EQI across all non-financial industries during the time period beginning just before the recession and ending at the present. Specifically, we look at data for the twelve months ended with Quarter 2 2007 through the twelve months ended with Quarter 2 2012.

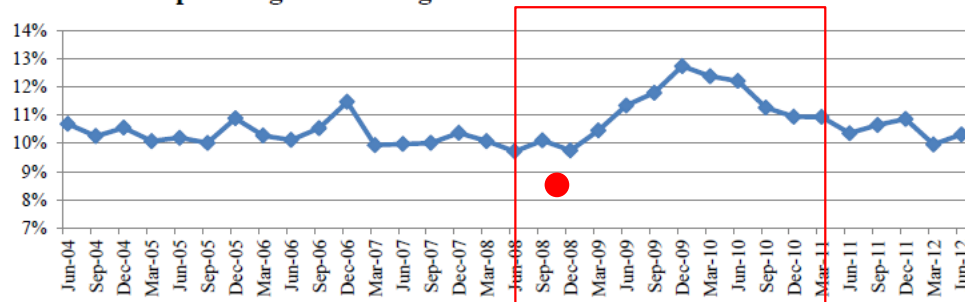
During this time period and as the chart below shows, we saw a large increase in the EQI ratio from 4.55% in June 2007 to 8.58% at its peak in December 2009. Driving EQI higher was an increase in operating cash margin and a decline in net margin.

Operating cash margin grew to 12.73% in December 2009 from 9.97% in June 2007 due primarily to lower operating receivable days, lower inventory days and higher accrual days. During that difficult period, companies conserved cash by carrying lower inventories and postponing the payment of operating expenses.

Accompanying the rise in operating cash margin was a decline in net margin, which during the same time period fell to 3.76% from 5.53% as the recession took its toll on profitability. The combination of rising operating cash margin and declining net margin served to increase EQI.

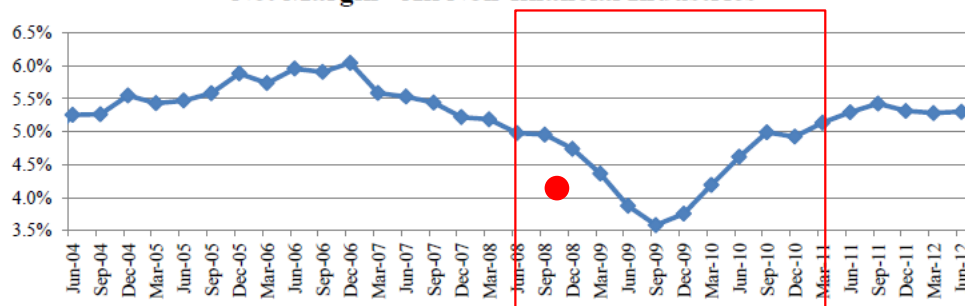
Since December 2009, operating cash margin has fallen to 10.31% in June 2012 while net margin has increased to 5.30%. In a recovering economy, both developments are expected. As a result, EQI has fallen to 4.56%, almost exactly where it was five years earlier.

Operating Cash Margin - All Non-financial Industries



Operating Cash Margin = Sustainable Operating Cash Flow / Revenue

Net Margin - All Non-financial Industries



Net Margin = Sustainable Net Earnings / Revenue



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	t-2	t-1	t
Sales Revenues	88,00	100,00	120,00
EBITDA	17,60	20,00	24,00
D/A	-3,50	-4,00	-5,00
Oper. Income	14,10	16,00	19,00
CAPEX	N/A	4,00	6,00
ΔNOWC	N/A	3,00	5,00
NOWC	22,00	25,00	30,00
NOFA	50,00	50,00	51,00
NOA	72,00	75,00	81,00
EBITDA	17,60	20,00	24,00
ΔNOWC	N/A	-3,00	-5,00
OCF	N/A	17,00	19,00
CAPEX	N/A	-4,00	-6,00
OFCF	N/A	13,00	13,00
NOWC / Revenues	25,0%	25,0%	25,0%
CAPEX / Revenues	N/A	4,0%	5,0%
ROS	16,02%	16,00%	15,83%
D/A / Revenues	-3,98%	-4,00%	-4,17%
EBITDA Margin	20,00%	20,00%	20,00%
ΔNOWC / Revenues	N/A	3,00%	4,17%
OCF / Revenues	N/A	17,00%	15,83%
CAPEX / Revenues	N/A	4,00%	5,00%
OFCF / Revenues	N/A	13,00%	10,83%

Introduction:

We have seen that OFCF is a fundamental metric in order to understand how good is a business in generating cash from its operating income and assets.

Two ratios are, therefore, very important:

- **OCF / Revenues** (cash profitability of sales)
- **CAPEX / Revenues** (the CAPEX-to-Revenues ratio that measures the relevance of net investment given the level of operating activity, i.e., sales revenues).

Given the significance of these two ratios, even the ratio

OFCF / Revenues

is a key metric (as we have seen for the two companies from the *theme parks* industry).

Now, firstly, read carefully the data on the left side and then start with the animation!

	t-2	t-1	t
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OCF / Revenues	N/A	17,00%	15,83%
CAPEX / Revenues	N/A	4,00%	5,00%
OFCF / Revenues	N/A	13,00%	10,83%

$$\frac{\Delta \text{NOWC}}{\text{Revenues}} = \frac{\text{NOWC}_{t-1}}{\text{Reven}_{t-1}} \times \frac{g_{\text{NOWC}}}{g_{\text{Rev}}} \times \frac{g_{\text{Rev}}}{1 + g_{\text{Rev}}}$$

t-1	3,00%	=	25,00%	x	1,00	x	12,00%
t	4,17%	=	25,00%	x	1,00	x	16,67%

$$g_{\text{Rev}}^{t-1} = (100-88)/88 = 13,6\%$$

$$g_{\text{Rev}}^t = (120-100)/100 = 20,0\%$$

	Oper. Inc.	+	D/A
	Revenues		Revenues
t-1	16,00%	t-1	4,00%
t	15,83%	t	4,17%

	EBITDA	-	ΔNOWC
	Revenues		Revenues
t-1	20,00%	t-1	-3,00%
t	20,00%	t	-4,17%

	CAPEX	X	D/A
	D/A		Revenues
t-1	1,00	x	4,00%
t	1,20	x	4,17%

	OCF	-	CAPEX
	Revenues		Revenues
t-1	17,00%	t-1	4,00%
t	15,83%	t	5,00%

The ratio OCF / Revenues is worse (from 17,00% to 15,83%) even if:

- EBITDA Margin is stable (20%)
- NOWC / Revenues is stable (25%), i.e.
- $g_{\text{NOWC}} = g_{\text{Revenues}}$ (same growth rate)

WHY?

Because the ratio $\Delta \text{NOWC} / \text{Revenues}$ is also influenced by the ratio: $g_{\text{Revenues}} / 1 + g_{\text{Revenues}}$

All things being equal, the higher the growth in revenues ("acceleration of growth"), the higher the impact of the increase in NOWC on the OCF/Revenues ratio (growth rate from 13,6% to 20%)

	OFCF
	Revenues
t-1	13,00%
t	10,83%

IS (ADVANCED COURSE)

There are two relevant metrics: **Operating Cash Flow** and **Operating Free Cash Flow** (after subtracting **CAPEX**).

1. When you analyze the Operating Cash Flow you must focus your attention to the two components:

- a) “Monetary” (and recurring) Operating Income (EBITDA)
- b) Change in (recurring) Net Operating Working Capital

Therefore, there are two “normalization” levels:

- a) the **difference between monetary revenues and monetary expenses**: we apply the same approach we use in analyzing the *usual/recurring/sustainable core operating income*.
- b) The **change in net operating working capital**: the objective is to measure the *expected recurring change in net operating working capital items*, according to the growth profile of the company (expected “normal” change in A/R, inventory, A/P, other operating current assets and operating liabilities):
 - ✓ This change is expected to be 0 when the company’s trend is stable (no growth, no decline);
 - ✓ This change is a specific percentage of the change in sales revenues if the company’s trend is positive (growth) or negative (decline): **remember the ratio NOWC/Net Sales!**

2. When you analyze the CAPEX, you must carefully understand which is the **recurring amount for maintenance investments** (analyzing the two ratios *Capex-to-Revenues* and *Capex-to-Depreciation and Amortization*)



In a trend analysis of the «reformulated» cash flow statement it's crucial to use charts, specifically the «**Stacked Bar Charts**»

Using the following rules:

- ☐ Positive cash flows above the horizontal axis and Negative cash flows below the horizontal axis
- ☐ Separating unusual cash flows and the cash flows with related parties (change in intercompany NOWC; intercompany financial transactions)

Lets' see some examples

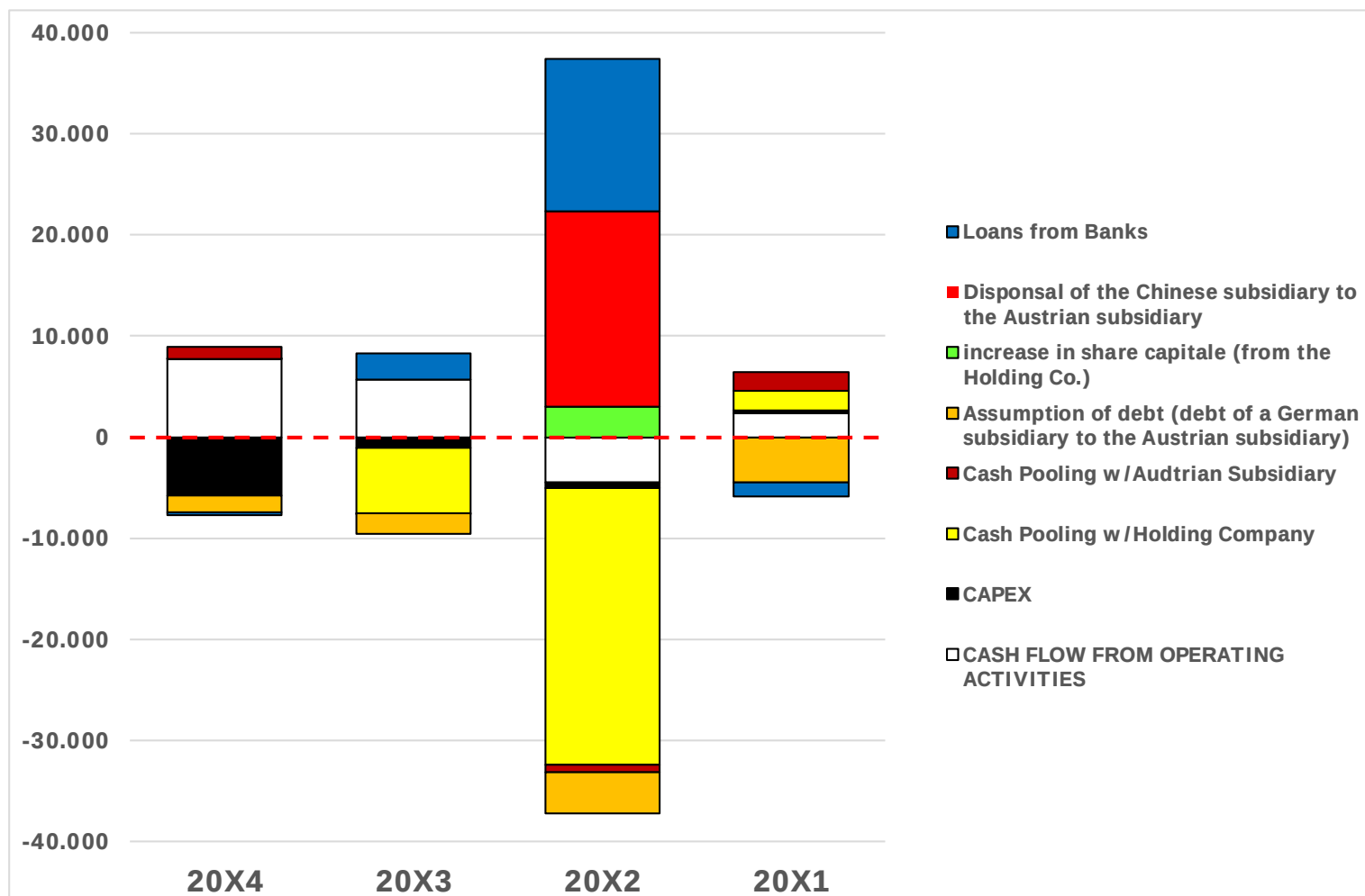


	20X4	20X3	20X2	20X1
CASH FLOW FROM OPERATING ACTIVITIES	7.735	5.697	-4.477	2.343
CAPEX	-5.764	-1.045	-552	289
Cash Pooling w/Holding Company	0	-6.448	-27.377	1.997
Cash Pooling w/Audtrian Subsidiary	1.234	-68	-751	1.809
Assumption of debt (debt of a German subsidiary to the Austrian subsidiary)	-1.660	-2.030	-4.040	-4.484
increase in share capitale (from the Holding Co.)	0	0	3.000	0
Disposnal of the Chinese subsidiary to the Austrian subsidiary	0	0	19.336	0
Loans from Banks	-303	2.581	15.079	-1.397
NET CHANGE IN CASH	1.242	-1.313	219	557
EBITDA	-7.267	-3.424	-2.994	-1.156
Net financial expenses	-1.744	-1.248	-1.677	-2.474
Unrealized gains/losses on financial assets	0	0	0	0
Extraordinary Items	-236	-1.115	144	-164
Income taxes	-110	-380	-350	-342
"Monetary" Net Income (Loss)	-9.357	-6.166	-4.876	-4.137
Change in NOWC with third parties	19.928	7.665	2.595	7.564
Change in NOWC with related parties	-3.298	5.712	104	1.108
Change in Tax Assets and Liabilities	784	-1.085	-1.137	290
Change in Provision Liabilities	-323	-429	-1.162	-2.483
Change in Assets and Liabilities	17.092	11.863	400	6.480
CASH FLOW FROM OPERATING ACTIVITIES	7.735	5.697	-4.477	2.343





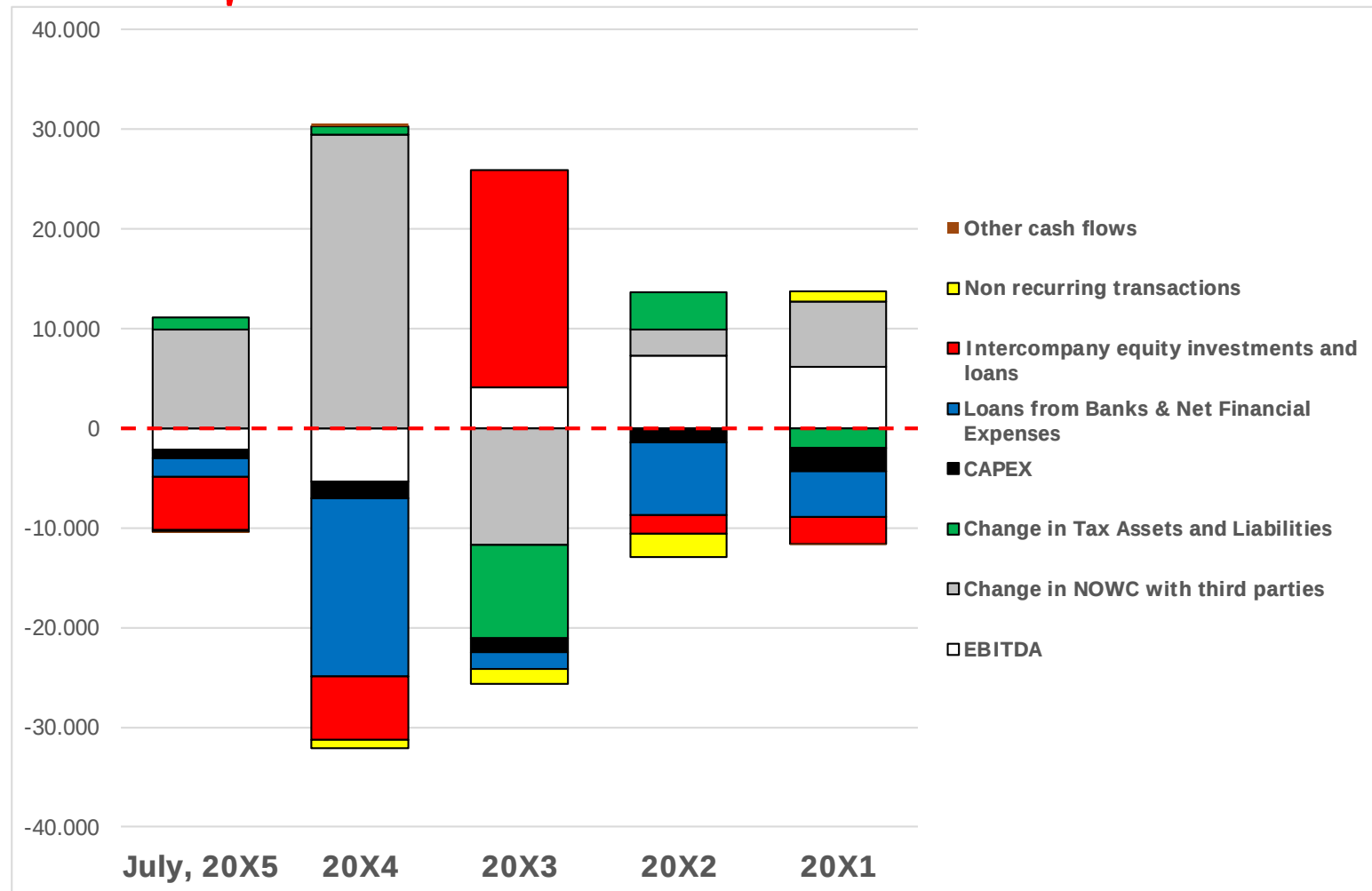
Insolvency in 20X5...



	July, 20X5	20X4	20X3	20X2	20X1
EBITDA	-2.180	-5.303	4.074	7.301	6.184
Change in NOWC with third parties	9.915	29.507	-11.660	2.592	6.519
Change in Tax Assets and Liabilities	1.223	822	-9.357	3.795	-1.999
CAPEX	-782	-1.732	-1.431	-1.364	-2.345
Loans from Banks & Net Financial Expenses	-1.905	-17.821	-1.735	-7.372	-4.567
Intercompany equity investments and loans	-5.308	-6.419	21.883	-1.865	-2.708
Non recurring transactions	-212	-819	-1.501	-2.287	1.039
Other cash flows	-2	220	0	0	-1
NET CHANGE IN CASH	751	-1.766	273	800	2.123



 **Insolvency in 20X5...**



	20X6	20X5	20X4	20X3	20X2	20X1
Financial Assets (trading)	-534	572	-1.557	924	-1.074	-1.771
Loans to Customers (net position)	482	-3.270	565	556	3.914	-368
Bondholders	-1.130	-2.085	64	-921	-984	325
Loans to/from Banks (net position)	-292	4.777	-1.945	1.158	-1.711	1.684
Other assets and liabilities	313	-158	245	-186	-69	10
Cash Flow from Operating Activities	-369	103	-109	153	178	139
Equity Investments and CAPEX (and related gains/losses)	-142	29	-73	18	-23	-20
Issuance of shares and Purchases/Sales of Own Shares	1.660	14	618	274	-20	27
NET CHANGE IN CASH	-13	-17	-2.191	1.977	212	26





Insolvency in 20X7...

